Witchayut Tangwongkitsiri

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Portfolio: https://knotnot.github.io/portfolio/

Summary

Bachelor's degree in electrical engineering, *Chulalongkorn University* (2021 - 2025)

With academic foundations in **mathematics**, **control systems**, and wireless **communication**, as well as exposure to **cybersecurity** concepts. While my degree focused on communication and control systems, I developed a strong interest in **data engineering**, which led me to pursue related projects and coursework in this field.

Relevant coursework :

- Data Science Practicum
- AI Engineering
- Computer Programming

Experience

IOT Development Intern

Electrical Engineering Technology for Agricultural Resources (EETAR) (June 2024 – August 2024)

- **Developed IoT-based systems** with Arduino IDE to automate agricultural tasks, collect data, and integrate with InfluxDB for real-time monitoring.

Math Tutor

(2021 - present)

- Taught at institutions such as Mainstairs and NCU Tutoring Center
- Tutored students in mathematics (elementary to high school) through group and one-on-one sessions.

Project

- Development Cyber Security Testbed – Mininet Wifi
  - **Development of network security testbed** for abnormal traffic data detection in cyber-attack analysis using machine learning: **Server module**
  - **Co-author** of a research paper accepted for publication in IEEE 2025 ISCIT: **"Design of Software-Defined Wireless Network for Machine Learning-Driven DDoS Detection and Tiered Response."** (Project repo: https://github.com/knotnot/senior-project_server-module)
- Creating data pipeline ETL & automate to show top 10 search result from user keywords
  - **Developed an ETL pipeline** in Python to extract data from the TripAdvisor API, transform and clean the data, and load it into a MySQL database running in a Docker container. **Automated the workflow** using Apache Airflow, also deployed in a separate Docker container. (Project repo: https://github.com/knotnot/ETL-data-pipeline)
- Integrating machine learning into a Flask-based web application
  - **Developed a machine learning model** using fertilizer application and weather data. **Deployed the model with Flask** to create a web interface for practical use.

Additional & Skills

- **Language:** Thai (Native), English (Fluent), TOEIC score: 805/990
- **Technical Skills:** Python, GCP, AWS, Databricks, SQL, HTML, CSS, Docker container, Power BI, Ubuntu (Linux), Git, MySQL, Excel, Matlab, Apache Airflow
- **Activities:**
  - MC for Electrical engineering faculty
  - CUEE-SDG: Community Service: Electrical Repair at Rural School
- **Interests:** Movies, Board games, Cloud technologies, Investment, Tennis, AI